

CHAPTER 3: FUNCTIONAL REQUIREMENTS

3.1 User & Access Management

Table 3-1: User & Access Management Functional Requirements

ID	Description	Priority	Source	Status
FR-001	The system shall support multiple user roles: Customer, Operator, Driver, Staff, and Admin, with access controlled using a role-permission matrix.	High	System	Proposed
FR-002	User accounts shall be created, updated, and deactivated by the Admin.	High	Admin	Proposed
FR-003	Role-based permissions shall be configurable and enforceable across all system modules.	High	Admin	Proposed
FR-004	Users shall be authenticated using a valid email and password.	High	System	Proposed
FR-005	Users shall be redirected to role-specific dashboards after successful authentication.	High	System	Proposed
FR-006	User accounts shall be locked after five consecutive failed login attempts.	High	System	Proposed
FR-007	Password reset shall be performed via a time-limited email link.	High	System	Proposed
FR-008	User sessions shall be terminated after 30 minutes of inactivity.	Medium	System	Proposed

3.2 Customer Registration & Profile Management

Table 3-2: Customer Registration Functional Requirements

ID	Description	Priority	Source	Status
FR-009	Customers shall be able to register by providing their name, email, phone number, and password.	High	Customer	Proposed
FR-010	Customer profiles shall be updatable including contact details and preferences.	High	Customer	Proposed
FR-011	Customer booking history shall be viewable from the customer profile.	High	Customer	Proposed

3.3 Bus Ticket Booking

Table 3-3: Bus Ticket Booking Functional Requirements

ID	Description	Priority	Source	Status
FR-012	Customers shall be able to search for available buses by selecting origin, destination, and travel date.	High	Customer	Proposed
FR-013	The system shall display available seats in real-time for each bus route.	High	System	Proposed
FR-014	Customers shall be able to select one or more seats and confirm a booking.	High	Customer	Proposed
FR-015	The system shall prevent double-booking of the same seat.	High	System	Proposed
FR-016	Booking confirmation shall be sent to the customer via email or SMS upon successful reservation.	High	System	Proposed
FR-017	Customers shall be able to cancel a bus booking within the permitted cancellation window.	High	Customer	Proposed
FR-018	Booking status shall be maintained as Confirmed, Cancelled, or Completed.	High	System	Proposed
FR-019	An incomplete bus booking shall be automatically cancelled by the system if payment or confirmation is not completed within 15 minutes of initiation.	High	System	Proposed

3.4 Route & Schedule Management

Table 3-4: Route & Schedule Management Functional Requirements

ID	Description	Priority	Source	Status
FR-020	Admin shall be able to create, update, and deactivate bus routes including origin, destination, and stops.	High	Admin	Proposed
FR-021	Admin shall be able to configure schedules including departure times, days of operation, and seat capacity.	High	Admin	Proposed
FR-022	The system shall automatically update seat availability when bookings are made or cancelled.	High	System	Proposed
FR-023	Admin shall be able to set and adjust ticket pricing per route.	High	Admin	Proposed

3.5 Airport Transfer Booking

Table 3-5: Airport Transfer Booking Functional Requirements

ID	Description	Priority	Source	Status
FR-024	Customers shall be able to book airport pickup and drop-off services by specifying date, time, location, and flight details.	High	Customer	Proposed
FR-025	The system shall display available vehicles and estimated transfer pricing for airport transfers.	High	System	Proposed
FR-026	Airport transfer booking confirmation shall be sent to the customer upon successful reservation.	High	System	Proposed
FR-027	Customers shall be able to cancel an airport transfer booking.	High	Customer	Proposed
FR-028	Admin shall be able to assign a driver and vehicle to confirmed airport transfer bookings.	High	Admin	Proposed

3.6 Tour Package Customization

Table 3-6: Tour Package Functional Requirements

ID	Description	Priority	Source	Status
FR-029	Customers shall be able to create customized Sri Lankan tour packages by selecting from a list of available tourist destinations.	High	Customer	Proposed
FR-030	The system shall automatically calculate tour package pricing based on selected destinations, vehicle type, and duration.	High	System	Proposed
FR-031	Customers shall be able to save, review, and confirm their customized tour packages before booking.	High	Customer	Proposed
FR-032	Tour package booking confirmation shall be generated and sent to the customer upon successful reservation.	High	System	Proposed
FR-033	Admin shall be able to manage the list of tourist destinations and associated pricing.	High	Admin	Proposed
FR-034	Customers shall be able to view and cancel their tour package bookings.	High	Customer	Proposed

3.7 Vehicle, Driver & Staff Management

Table 3-7: Vehicle, Driver & Staff Functional Requirements

ID	Description	Priority	Source	Status
FR-035	Admin shall be able to add, update, and deactivate vehicles including type, capacity, and registration details.	High	Admin	Proposed
FR-036	Admin shall be able to manage driver profiles including personal details, license information, and status.	High	Admin	Proposed
FR-037	Admin shall be able to assign drivers to specific routes, airport transfers, or tour bookings.	High	Admin	Proposed
FR-038	Staff schedules shall be manageable by the Operator.	Medium	Admin	Proposed
FR-039	Vehicle availability shall be updated automatically when assigned to a booking.	High	System	Proposed

3.8 Operator Branding & Pricing Control

Table 3-8: Operator Branding & Pricing Functional Requirements

ID	Description	Priority	Source	Status
FR-040	Admin shall be able to configure their brand identity including logo, company name, and contact details on the platform.	Medium	Admin	Proposed
FR-041	Admin shall be able to set and update pricing for bus routes, airport transfers, and tour packages independently.	High	Admin	Proposed
FR-042	Admin-specific pricing and branding shall be visible to customers during the booking process.	High	System	Proposed

3.9 Booking Management & Notifications

Table 3-9: Booking Management Functional Requirements

ID	Description	Priority	Source	Status
FR-043	Customers shall be able to view all their current and past bookings from their dashboard.	High	Customer	Proposed
FR-044	The system shall send real-time booking confirmation and cancellation notifications to customers via email or SMS.	High	System	Proposed
FR-045	Admin shall receive notifications when a new booking is made or cancelled.	High	System	Proposed
FR-046	Booking details shall include service type, date, time, passenger details, and payment status.	High	System	Proposed

3.10 Reporting & Analytics

Table 3-10: Reporting & Analytics Functional Requirements

ID	Description	Priority	Source	Status
FR-047	Admin shall be able to generate reports on bookings, revenue, and vehicle utilization.	High	Admin	Proposed
FR-048	The system shall provide an analytics dashboard showing booking trends, popular routes, and revenue summaries.	Medium	Admin	Proposed
FR-049	System-wide reports and audit logs shall be accessible to the Admin.	High	Admin	Proposed